



Calculating theoretical probabilities from Venn diagrams (two events)

Task A: Theoretical probabilities of combined events

1) Use the Venn diagram to find the probabilities.

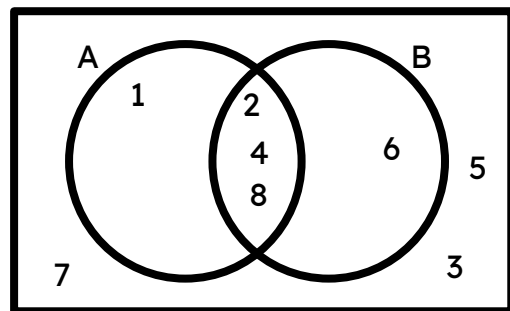
a) $P(A \text{ and } B) =$

b) $P(A \text{ and not } B) =$

c) $P(B \text{ and not } A) =$

d) $P(\text{not } A \text{ and not } B) =$

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$$



2) Use the Venn diagram to write the probabilities.

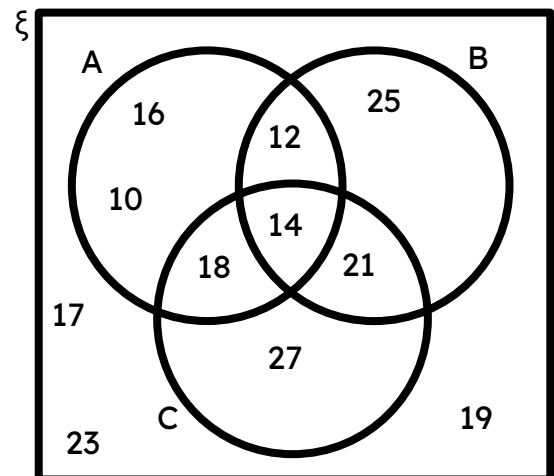
a) $P(A \text{ and } C) =$

b) $P(B \text{ and not } C) =$

c) $P(\text{not } C \text{ and not } A) =$

d) $P(A \text{ and not } B) =$

e) $P(\text{not } A \text{ and } B) =$



3) This Venn diagram shows the number of outcomes in each event.

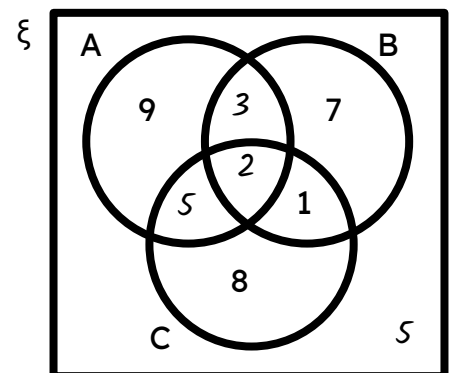
There are 40 outcomes in the sample space.

Complete the Venn diagram given these probabilities.

$P(\text{not } A) =$

$P(A \text{ and not } C) =$

$P(B) =$



Task B: Further combinations of events

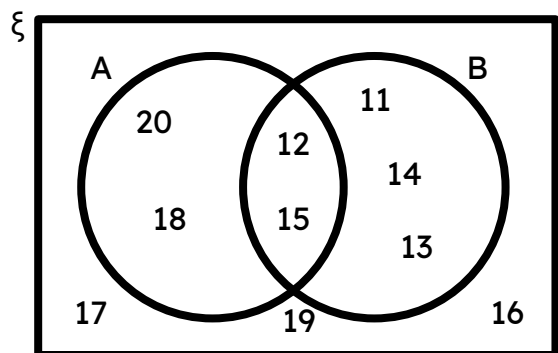
1) Use the Venn diagram to answer the following questions.

a) List the outcomes that are in event A.

b) List the outcomes that are not in event B.

c) Hence, list the outcomes that are in event A or not in event B.

d) Find $P(A \text{ or not } B)$



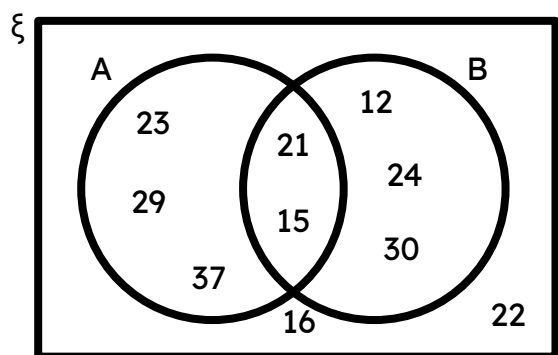
2) Use the Venn diagram to answer the following questions.

a) Find $P(A \text{ or } B)$

b) Find $P(A \text{ or not } B)$

c) Find $P(\text{not } A \text{ or not } B)$

d) Find $P(\text{only } A \text{ or only } B)$



3) Use the Venn diagram to answer the following questions.

a) Find $P(A \text{ or } B)$

b) Find $P(\text{not } A \text{ or } C)$

c) Find $P(\text{not } A \text{ or not } B)$

d) Find $P(\text{only } C \text{ or not } B)$

